



UI/UX DESIGN DOCUMENTATION

Intelligent Compound Maintenance & Home Services Platform
Three-Portal Product Design System — IEEE Academic
Submission

PROJECT NAME

Taska Platform

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UI/UX Documentation

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V1.0 — Final

SUBMISSION

Academic Evaluation

PORTALS COVERED

Resident · Technician · Admin

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Table of Contents

Overview & Design System

1. UI/UX Overview & Design System

1.1 Visual Identity & Color Palette

1.2 Typography & Iconography

1.3 Viewport Shell & Responsiveness

1.4 UI Component Catalog

Application Screens

2. Resident Portal Screens (taska-resident)

3. Technician Operations Screens (taska-technician)

4. Administrator Console Screens (admin-react)

User Flows & Architecture

5. User Flows & Interaction Journeys

6. Sitemap & Information Architecture

7. Routing Specifications & Screen-to-Feature Mapping

8. Responsive & Interaction Behavior

List of Screens

ID	Screen Name	Portal	Route
R-01	Resident Login	Resident	#/login
R-02	Compound Resident Registration	Resident	#/register
R-03	Two-Step OTP Verification	Resident	#/verify-otp
R-04	Resident Home & Service Catalog	Resident	#/resident-home
R-05	Create Service Request	Resident	#/create-order
R-06	Offers & Bidding Marketplace	Resident	#/order-offers
R-07	Live Order Tracking & Execution	Resident	#/order-tracking
R-08	Resident Profile & Settings	Resident	#/profile
R-09	Live Support Chat Widget	Resident	Floating Component
T-01	Technician Home & Radar	Technician	#/technician-home
T-02	Technician Job Execution & Tracking	Technician	#/tech-execution
A-01	Executive Operations Dashboard	Admin	/
A-02	Live Orders Kanban Board	Admin	/board
A-03	Support Ticket Desk	Admin	/support
A-04	Multi-Language Translation Manager	Admin	/translations

List of User Flows

#	Flow Name	Primary Actors
1	Resident Registration & Compound Verification	Resident, API, Mail Server, Database
2	Maintenance Request Creation & Reverse Bidding Marketplace	Resident, Technician, API, SignalR
3	Real-Time GPS Tracking & Job Execution Lifecycle	Technician, Resident, API, SignalR, Google Maps
4	In-Order Direct Real-Time Chat	Resident, Technician, API, SignalR
5	Customer Support Helpdesk Journey	Resident, Admin, API, SignalR
6	Administrative Operations & Real-Time Kanban Dispatch	Admin, API, SignalR

Section 1 — UI/UX Design System

Taska features a modern, mobile-first design language for resident and technician portals, complemented by a sleek, dark-themed operations dashboard for compound administrators.

1.1 Visual Identity & Color Palette

The color system is defined via CSS design tokens across both light and dark themes.

Light Theme — Resident & Technician Portals

Primary #385270	Primary Light #4A6B92	Primary Dark #273A50	Success #10B981
Warning #F59E0B	Danger #EF4444	Info #3B82F6	Accent Purple #8B5CF6
Accent Cyan #06B6D4	Background #F8F8F8	Surface #FFFFFF	Surface Alt #F1F5F9

Token Name	HEX Value	CSS Variable	Usage
Primary	#385270	--primary	Deep Slate Teal — Core Brand Accent
Primary Light	#4A6B92	--primary-light	Slate Blue Hover & Gradients
Primary Dark	#273A50	--primary-dark	Focused & Active State
Success	#10B981	--success	Emerald Green — Completed, Online, Paid
Warning	#F59E0B	--warning	Amber — Pending Orders, In Review
Danger	#EF4444	--danger	Coral Red — Cancelled, Failed, Delete
Info	#3B82F6	--info	Royal Blue — Accepted, Live Chat
Accent Purple	#8B5CF6	--accent-purple	Violet — In Progress Jobs, Analytics
Accent Cyan	#06B6D4	--accent-cyan	Cyan — Arrived Status, HVAC
Background	#F8F8F8	--background	Off-White / Canvas
Surface	#FFFFFF	--surface	Pure White Container
Surface Alt	#F1F5F9	--surface-alt	Slate Tinted Section Background
Border	#E2E8F0	--border	Subtle Dividers
Text Main	#0F172A	--text-main	Slate 900 — Main Heading Typography
Text Muted	#64748B	--text-muted	Slate 500 — Captions & Secondary Details

Admin Dashboard Dark Theme

Dark Base #0F172A	Card Surface #1A2235	Border Dark rgba(255, 255, 255, 0.07)
---	--	---

Token	Value	CSS Context	Usage
Dark Base	#0F172A	background-color	Slate 900 — Admin layout background
Card Background	#1A2235	--card-bg	Elevated card surface in dark mode
Dark Border	rgba(255, 255, 255, 0.07)	--border-dark	Subtle dividers on dark surfaces

1.2 Typography & Iconography

Category	Typeface	Usage
Primary Font	Inter, Outfit, system-ui sans-serif	All body text, headings, UI labels across all portals
Monospace Font	JetBrains Mono / SFMono	Order IDs, technical codes, GPS coordinates, API references

Icon Library	Used In	Example Icons
<code>lucide-react</code>	Resident & Technician apps	Bolt, Droplet, Snowflake, Paintbrush, Hammer, MapPin, Star, Send, X, Phone
Google Material Icons Round	Admin Dashboard	Navigation sidebar icons, KPI stat icons

1.3 Viewport Shell & Responsiveness

Application	Layout Approach	Implementation
Resident Portal Technician Portal	Mobile-first, centered phone shell on desktop	<code>ResponsiveLayout.tsx</code> — On desktop (>768px), centers content within max-width 480px smartphone frame container with drop shadows, mimicking native iOS/Android experience.
Admin Console	Full-width responsive desktop layout	Persistent collapsible sidebar, high-density data tables, and flexbox grid charts.

1.4 UI Component Catalog

Component	Found In	Description
<code>AppButton</code>	Resident / Tech	Primary, secondary, and danger button with loading spinner state and tactile feedback.
<code>AppInput</code>	Resident / Tech	Floating label input field with validation error indicators.
<code>Avatar</code>	Resident / Tech	Circular avatar rendering user initials or image URL with colored status badge.
<code>StarRating</code>	Resident / Admin	Interactive 1-5 clickable stars for review submission and static fractional rating display.
<code>TrackingMap</code>	Resident / Tech	Google Maps JavaScript API container with custom markers, polyline routing, and DirectionsRenderer.
<code>BottomSheet</code>	Resident / Tech	Framer Motion draggable bottom slide-up drawer for offers and bidding.
<code>AlertModal</code>	Resident / Tech	Global modal for critical notifications and confirmations.
<code>SupportChatWidget</code>	Resident	Floating expandable support chat widget.
<code>Sidebar</code>	Admin Console	Collapsible dark navigation drawer with Material Icons.
<code>StatCard</code>	Admin Console	Elevated metric card with colored icon, stat value, and comparison percentage.

Section 2 — Resident Portal Screens

The Resident Portal (`taska-resident` , port **5173**) uses `HashRouter` for PWA/mobile-web compatibility. All routes are prefixed with `#` .

R-01 Resident Login `#/login` Resident

PURPOSE

Authenticate registered residents into their compound maintenance session

TARGET USER

Gated compound residents

Main Components

- Header brand badge with Taska logo and tagline
- Role switcher toggle showing active "Resident" mode
- Email input field (`type="email"`)
- Password input field (`type="password"`) with show/hide toggle)
- Primary button: "Sign In"
- Secondary navigation link: "Don't have an account? Register"

Navigation & Validation

- On submit success → `#/resident-home`
- On register click → `#/register`
- Validation: Email format, password required (min 6 chars)

States & Error Handling

- Default · Loading spinner inside button · Error banner on invalid credentials
- 401 Unauthorized displays red toast: *"Invalid email or password."*

R-02 Compound Resident Registration `#/register` Resident

PURPOSE

Onboard residents by validating identity against their physical compound

TARGET USER

Prospective compound residents

Form Fields

1. Compound Code (e.g. `PH001` — required, validated against database)
2. Full Name
3. Email Address
4. Mobile Phone (`+20...`)
5. Apartment / Villa Code (e.g. `A1-402`)
6. Password & Confirmation

Validation Rules

- Compound code verified against active compound records
- Email unique validation · Phone unique validation · Password strength validation
- On success: Transfers email state to `#/verify-otp`
- Error: Toast if compound code does not exist

R-03 Two-Step OTP Verification #/verify-otp Resident

PURPOSE

Confirm resident email ownership before issuing access token

TARGET USER

Newly registered residents

Main Components

- Informational prompt displaying the destination email
- 6-digit numeric verification input box with auto-focus
- Countdown timer (10 minutes remaining)
- Primary button: "Verify & Enter"
- Resend OTP trigger link

States & Error Handling

- States: Ready · Verifying · Expired OTP state
- On success → receives JWT Bearer token → redirects to #/resident-home
- Error toast: "Invalid or expired OTP code."

R-04 Resident Home & Service Catalog #/resident-home Resident

PURPOSE

Primary hub for browsing compound services, seeing online technicians, and viewing active orders

TARGET USER

Compound residents

Main Components

- **Top App Header:** Compound name badge (Palm Hills October), search bar, notifications bell with unread count badge
- **Category Quick Grid:** Visual cards for Electricity, Plumbing, Air Conditioning, Gas, and Carpentry with base rates and icons
- **Interactive Compound Map:** Google Maps showing nearby verified technicians currently online within compound fences
- **Active Orders Card / Banner:** Highlights any ongoing order with real-time status chip and direct link to track
- **Notifications Slide-over Panel:** Drawer displaying live alerts (new bids, tech arrival)
- **Floating Support Chat Button:** Direct access to SupportChatWidget.tsx
- **Bottom Navigation Bar:** Tabs for Home, Active Orders, Profile

Navigation

- Tapping category card → #/create-order with category state
- Tapping active order card → #/order-tracking or #/order-offers

States

- Default · Loading skeleton · Empty active orders state
- Auto-reconnects to WebSocket channels if connection drops

R-05 Create Service Request <#/create-order> Resident

PURPOSE

Specify problem details, attach photo, select urgency and target coordinates

TARGET USER

Compound residents requesting repairs

Main Components

- Category header with base fee preview
- Problem description multi-line textarea with placeholder hints
- Service Urgency Level selector: *Standard* (20–30 min, Base price) · *Express* (10–15 min, + surcharge) · *Premium* (5–10 min, top-rated tech)
- Price suggestion input field (defaults to category base price)
- Payment method toggle: Cash / Visa Card
- Photo attachment button with camera icon, file picker, and thumbnail preview with remove button
- Interactive Google Maps mini-picker with crosshair pin for apartment coordinates
- Fixed bottom CTA button: "Submit Request"

Validation & Error Handling

- Description cannot be empty; checks `User::hasActiveOrder()` on server
- On submit → <#/order-offers> with newly created order ID
- If user already has active order → AlertModal: "You already have an active order. Please wait for it to complete."

R-06 Offers & Bidding Marketplace <#/order-offers> Resident

PURPOSE

Review competitive price quotes from online technicians and choose an offer

TARGET USER

Resident waiting for technician assignment

Main Components

- Gradient header displaying Order ID, Category, and Target Price
- Pulsing hourglass radar animation indicating live broadcasting to technicians
- Incoming Offers Feed: Technician avatar, full name & verified badge, 5-star rating (e.g. ★ 4.8), completed jobs count, quoted EGP amount, explanatory notes quote bubble, "Accept Offer" button

Navigation & States

- Once offer accepted → <#/order-tracking>
- If technician accepts at base price → Auto-transitions via WebSocket listener
- States: Waiting for bids (radar) · Multiple bids populated · Bid accepting loader
- Prevents double acceptance if offer expires or order was cancelled

R-07 Live Order Tracking & Execution #/order-tracking Resident

PURPOSE

Monitor technician movement on map, communicate in real time, settle payment, and rate service

TARGET USER

Resident awaiting and receiving service

Main Components

- **Full Google Map View:** Resident apartment pin, technician vehicle marker, animated DirectionsRenderer route
- **Trip Information Pill:** Live distance (e.g., 450 m) and duration (e.g., 3 mins)
- **Technician Profile Bar:** Avatar, name, phone dialer button, chat toggle with unread badge
- **Status Stepper Bar:** Horizontal timeline: Accepted → Arrived → In Progress → Completed
- **Live In-Order Chat Modal:** Flyout drawer for two-way messaging with technician
- **Cancel Order Button & Modal:** Cancellation with dropdown reason before work starts
- **Payment Settlement Modal:** Cash confirmation or simulated Visa card payment
- **5-Star Rating Modal:** 5 clickable stars, text feedback textarea, and submit button

Order Status States

Status	UI Behavior
accepted	Shows technician moving towards compound/apartment on map
arrived	Green banner: "Technician has arrived at your door!"
in_progress	Status chip: "Work in progress"
completed	Triggers payment & rating modals; navigates to home on completion

SignalR connection reconnects automatically via @microsoft/signalr with exponential backoff on network interruption.

R-08 Resident Profile & Settings #/profile Resident

PURPOSE

Manage account details, compound affiliation, notification toggles, language preferences, and logout

TARGET USER

Resident

Main Components

- Hero Profile Card with user initials avatar, phone number, and apartment code
- Metric counters: Total Services Completed, Rating, Compound Code (PH001)
- Settings list: Push notifications toggle · Language selector (English / العربية with RTL toggle) · Dark/Light mode toggle · Terms and Support link
- Destructive button: "Sign Out"

Navigation

- Sign out → Clears localStorage → Routes to #/login

R-09

Live Support Chat Widget

Floating Component – SupportChatWidget.tsx

Resident

PURPOSE

On-demand customer service directly connected to compound administrators

TARGET USER

Resident experiencing issues or needing help

Main Components

- Floating circular action button with message icon and unread badge
- Expandable modal with ticket subject, linked order selector, and initial message input
- Chat thread rendering: system notices, resident messages (right/blue), admin replies (left/dark)
- Send message input field
- Resolution satisfaction rating prompt (1–5 stars) when ticket is closed by admin

States

- Minimized bubble · Launcher form · Active conversation · Resolved/rating state

Section 3 — Technician Operations Screens

The Technician Portal (`taska-technician` , port **5174**) uses `HashRouter` .

T-01 Technician Home & Radar

`#/technician-home`

Technician

PURPOSE

Control duty status, view earnings, and discover maintenance requests in the compound

TARGET USER

Stationed compound technicians

Main Components

- **Duty Status Bar:** Compound badge (Palm Hills October), Online/Offline toggle switch with pulsing status dot
- **Daily Performance Counters:** Today's Earnings (e.g. `EGP 450`), Completed Jobs Count, Average Rating (e.g. `★ 4.9`)
- **Compound Order Radar Feed:** Cards for all pending requests — Category tag with color chip, Resident apartment code (e.g. `Villa 12B`), Problem description, Proposed price badge, Two action triggers:
 1. "Accept at EGP {Price}" — Direct instant acceptance
 2. "Place Custom Bid" — Opens bidding drawer
- **Bidding Bottom Sheet Modal:** Numeric input for custom offer amount, notes textarea, and submit button

Navigation & Validation

- On order assignment → immediately navigates to `#/tech-execution`
- Bidding amount must be greater than 0; cannot accept if already has active order
- States: Offline (radar dormant) · Online searching · Active job banner

T-02 Technician Job Execution & Tracking

`#/tech-execution`

Technician

PURPOSE

Navigate to resident's unit, stream GPS, communicate with resident, and progress job status

TARGET USER

Active technician on duty

Main Components

- **Full Google Navigation Map:** Route from technician's live GPS position to resident's apartment
- **Turn-by-turn ETA Pill:** Dynamic distance and arrival time
- **Resident Coordination Card:** Resident name, apartment code, direct phone call button (`tel:` link), in-app chat toggle with unread counter
- **Lifecycle Action CTA Button:**
 - State 1 (`accepted`): "I Have Arrived" — sets status to `arrived`
 - State 2 (`arrived`): "Start Working" — sets status to `in_progress`
 - State 3 (`in_progress`): "Complete Job" — sets status to `completed`
- **Job Finished Confirmation Card:** Completed service summary and earnings earned

Navigation & States

- On completion → returns to `#/technician-home`
- States: Step 1 (En route) · Step 2 (At customer door) · Step 3 (Working) · Step 4 (Complete)

Section 4 — Administrator Console Screens

The Admin Console (`admin-react` , port **5175**) uses `BrowserRouter` with standard URL paths. Full-width dark-themed responsive layout with persistent collapsible sidebar.

A-01 Executive Operations Dashboard / Admin

PURPOSE

High-level operational oversight across all compounds

TARGET USER

Platform administrators and compound facility managers

Main Components

- **Top KPI Metric Cards:** Total Registered Residents · Active Technicians (with online count) · Total Orders & Pending Orders · Total Paid Revenue (in EGP)
- **Live WebSocket Activity Indicator:** Green pulsing dot indicating live sync
- **Monthly Revenue Area Chart (Recharts):** Trajectory of paid revenue over time
- **Order Status Distribution Pie Chart:** Proportional breakdown of pending, active, completed, and cancelled requests
- **Recent Orders Table:** Top 10 latest requests with resident name, category, price, and status chips

A-02 Live Orders Kanban Board /board Admin

PURPOSE

Visual drag-and-drop operational board tracking all compound service requests

TARGET USER

Dispatchers and compound operations managers

Kanban Columns

Column	Color	Meaning
Pending	Amber	Unassigned requests awaiting bids/acceptance
Accepted	Blue	Technician assigned, en route to apartment
In Progress	Purple	Work underway on-site
Completed	Green	Work finalized, invoice settled

Features

- Draggable Order Cards: Order ID, Category icon, Resident name, Technician name, Price, timestamp
- WebSocket auto-sort: Cards automatically move to new column when technicians update status
- Click-to-inspect modal: Full order audit trail, status history, and payment details

A-03 Support Ticket Desk /support Admin

PURPOSE

Resolve resident support cases in real-time

TARGET USER

Administrative support agents

Main Components

- **Left Pane:** Ticket queue listing open cases with resident name, order link, status, and creation time
- **Right Pane:** Active chat dialogue showing full message history
- Administrative message composer with send button
- "Resolve & Close Ticket" action button

A-04

Multi-Language Translation Manager

/translations

Admin

PURPOSE

Manage UI text strings for English and Arabic dynamically

TARGET USER

Platform administrators

Main Components

- Language tab switcher (English vs العربية)
- Searchable key-value table: Translation Key (code reference) and Translated Value (display string)
- Inline input fields for modifying text
- Save Changes primary button (syncs changes to database)
- "Add Language" modal for provisioning additional locales

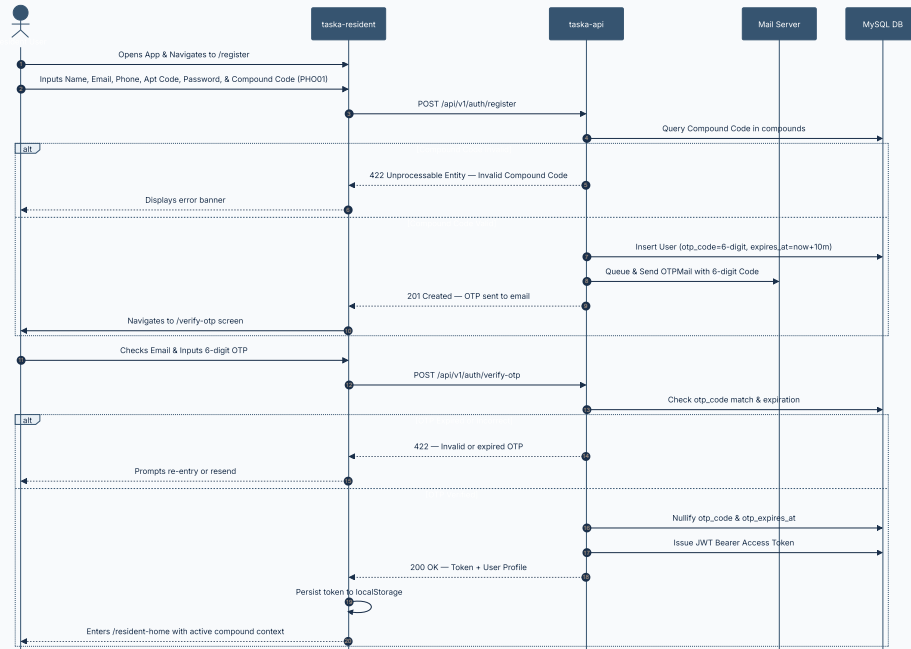
Section 5 — User Flows & Interaction Journeys

All user flows are derived directly from the application code, route definitions, and WebSocket event sequences.

1 Resident Registration & Compound Verification Flow

Actors: Resident User, taska-resident App, taska-api, Mail Server, MySQL DB

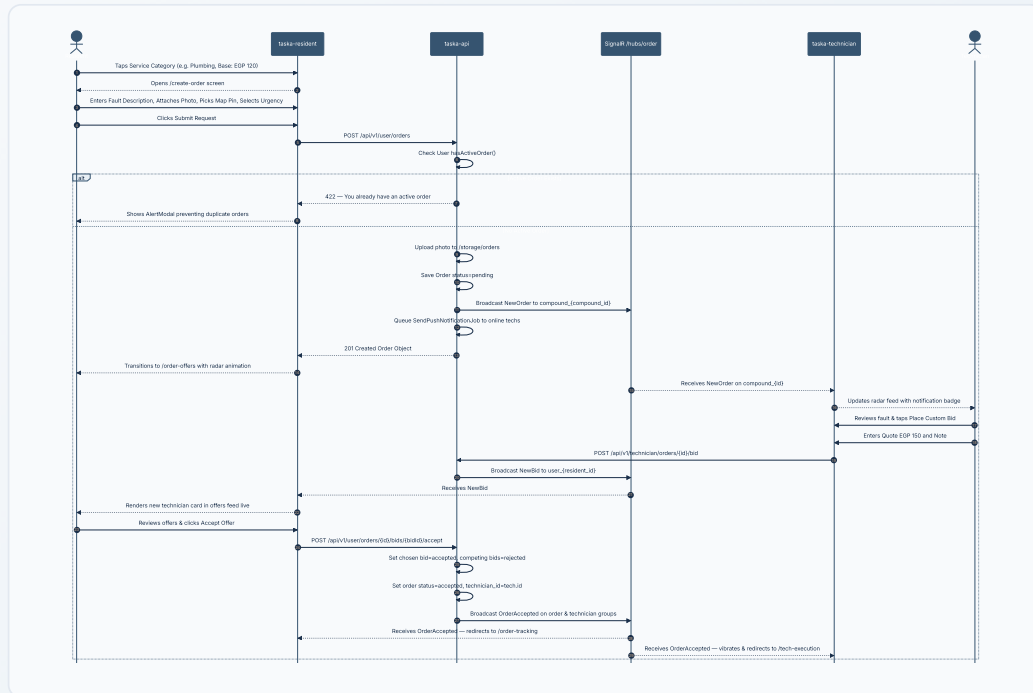
This flow validates that only physical residents of authorized gated compounds can access the marketplace.



2 Maintenance Request Creation & Reverse Bidding Marketplace Flow

Actors: Resident, taska-resident App, taska-api (ASP.NET Core), ASP.NET Core SignalR (/hubs/order), taska-technician App, Technician

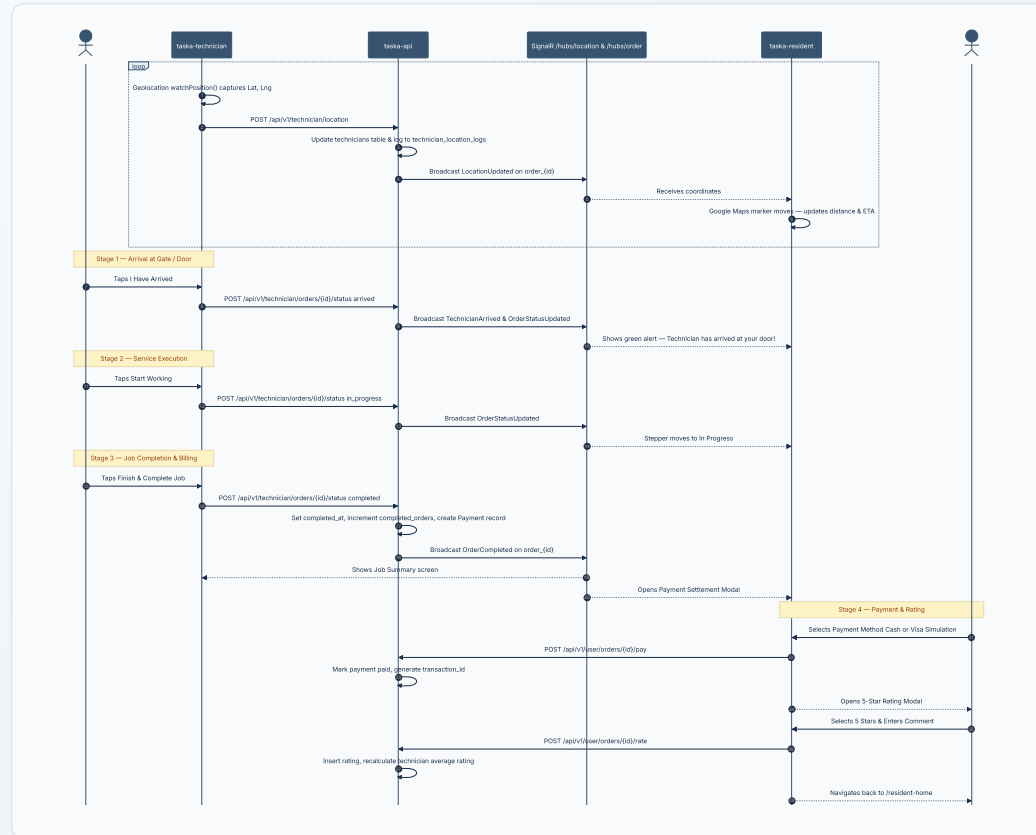
This journey illustrates the reverse bidding model where compound technicians submit monetary offers on a resident's request.



3 Real-Time GPS Tracking & Job Execution Lifecycle Flow

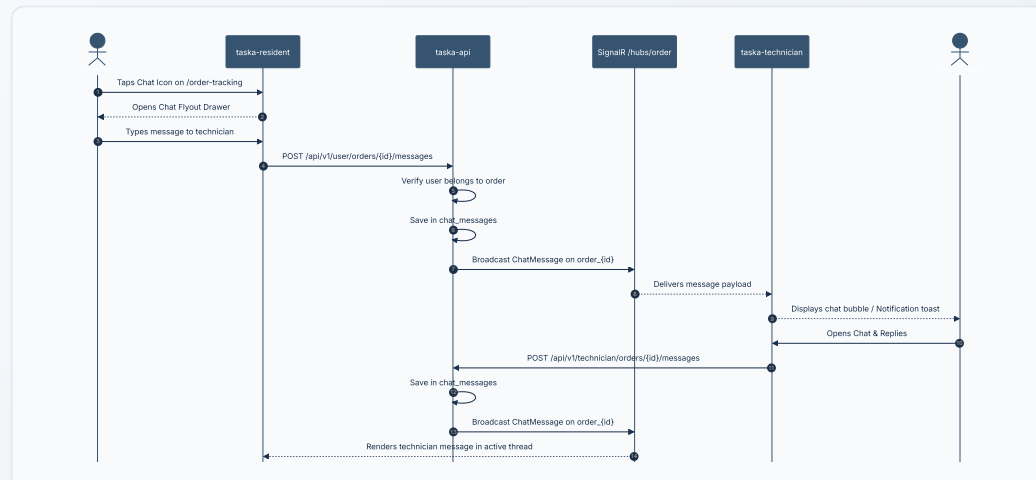
Actors: Technician, taska-technician, taska-api (ASP.NET Core), SignalR (/hubs/location & /hubs/order), taska-resident, Resident

This flow tracks physical movement and operational status progression from departure to job completion.



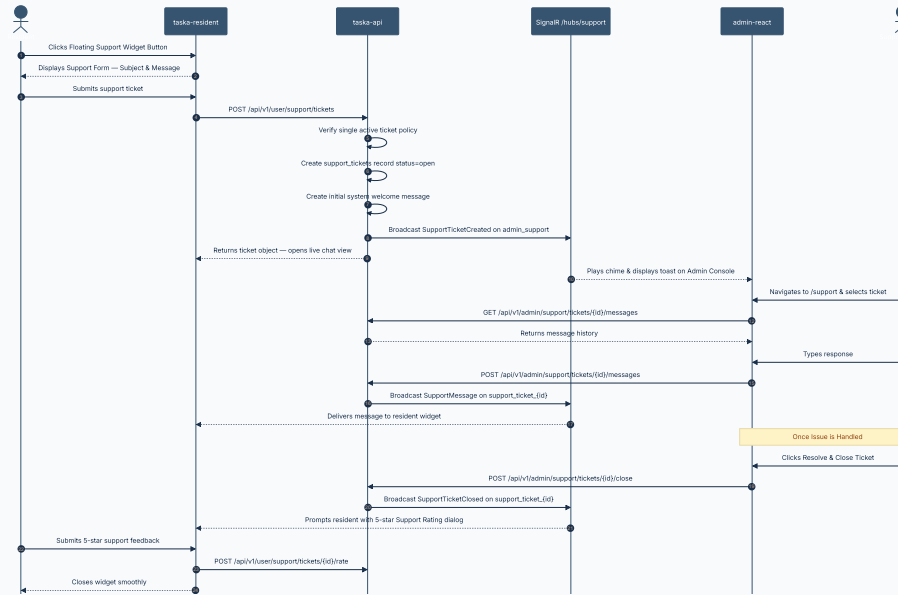
4 In-Order Direct Real-Time Chat Flow

Actors: Resident, taska-resident, taska-api (ASP.NET Core), SignalR (/hubs/order), taska-technician, Technician



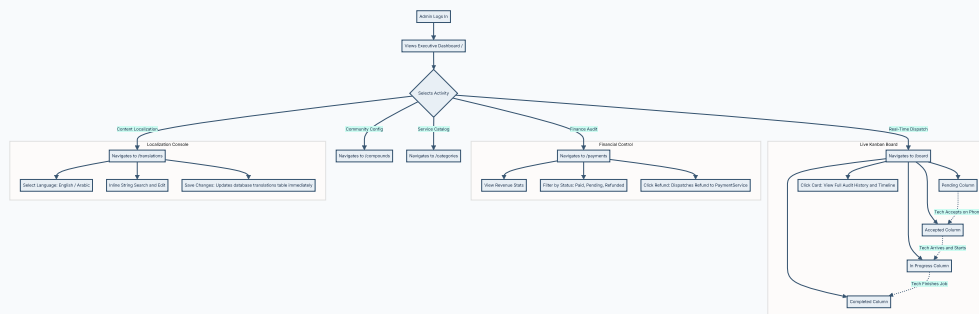
5 Customer Support Helpdesk Journey

Actors: Resident, taska-resident, taska-api, SignalR (/hubs/support), admin-react, Support Admin



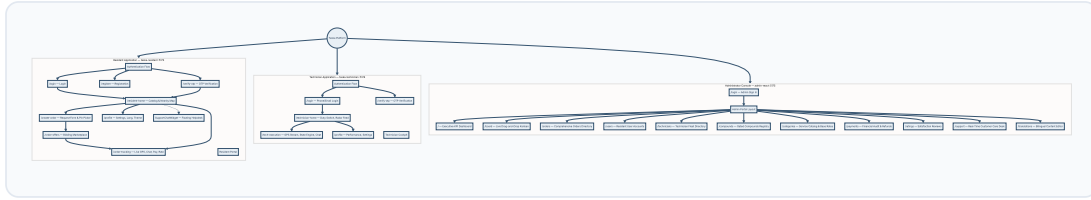
6 Administrative Operations & Real-Time Kanban Dispatch

Actors: Admin, API, SignalR, Kanban Board



Section 6 — Sitemap & Information Architecture

Complete information architecture, page hierarchy, URL routing schemas, and screen-to-feature mapping across the three client applications of Taska.



Section 7 — Routing Specifications & Screen-to-Feature Mapping

7.1 Resident Portal — taska-resident (Port 5173)

Router Type: HashRouter (PWA / Mobile Web Universal Compatibility)

Route Path	Page Component	Access	Functional Capabilities
<code>#/login</code>	<code>Login.tsx</code>	Public	Email and password authentication; navigation to register
<code>#/register</code>	<code>Register.tsx</code>	Public	Compound code validation, resident profile creation, triggers OTP
<code>#/verify-otp</code>	<code>VerifyOtp.tsx</code>	Public	6-digit numeric OTP validation; issues Bearer token
<code>#/resident-home</code>	<code>Home.tsx</code>	Auth Resident	Category selector, online technician map, active order card, alerts
<code>#/create-order</code>	<code>CreateOrder.tsx</code>	Auth Resident	Problem description, photo upload, map pin picking, urgency selector
<code>#/order-offers</code>	<code>OrderOffers.tsx</code>	Auth Resident	Live bidding marketplace, technician comparisons, bid acceptance
<code>#/order-tracking</code>	<code>OrderTracking.tsx</code>	Auth Resident	Live GPS vehicle tracking, in-order chat, payment modal, 5-star rating
<code>#/profile</code>	<code>Profile.tsx</code>	Auth Resident	Account information, bilingual toggle (AR/EN), theme toggle, sign out
<i>(Floating)</i>	<code>SupportChatWidget.tsx</code>	Auth Resident	In-app support ticketing and direct admin live messaging

7.2 Technician Field Portal — taska-technician (Port 5174)

Router Type: HashRouter

Route Path	Page Component	Access	Functional Capabilities
<code>#/login</code>	<code>Login.tsx</code>	Public	Login via mobile phone number or email address + password
<code>#/verify-otp</code>	<code>VerifyOtp.tsx</code>	Public	OTP verification screen
<code>#/technician-home</code>	<code>TechHome.tsx</code>	Auth Tech	Duty online/offline switch, daily earnings, compound request radar feed, bidding drawer
<code>#/tech-execution</code>	<code>TechExecution.tsx</code>	Auth Tech	Live GPS streaming, turn-by-turn routing, customer details, in-order chat, status progression buttons
<code>#/profile</code>	<code>Profile.tsx</code>	Auth Tech	Trade specialization badge, rating stars, completed jobs history, logout

7.3 Administrator Console — admin-react (Port 5175)

Router Type: BrowserRouter

Route Path	Page Component	Access	Functional Capabilities
<code>/login</code>	<code>Login.jsx</code>	Public	Administrator credentials authentication
<code>/</code>	<code>Dashboard.jsx</code>	Auth Admin	KPI stat cards, Recharts revenue trajectories, order status pie chart
<code>/board</code>	<code>OrdersBoard.jsx</code>	Auth Admin	Real-time drag-and-drop Kanban dispatch board across 4 status lanes

Route Path	Page Component	Access	Functional Capabilities
/orders	Orders.jsx	Auth Admin	Searchable orders data table, compound filters, status history modal
/users	Users.jsx	Auth Admin	Resident directory, apartment code registry, deletion action
/technicians	Technicians.jsx	Auth Admin	Technician fleet manager, provisioning modal, duty status filters
/compounds	Compounds.jsx	Auth Admin	Gated community registry, compound code CRUD, address and city fields
/categories	Categories.jsx	Auth Admin	Service catalog grid, base price editor, icon and color assignment
/payments	Payments.jsx	Auth Admin	Transaction ledger, revenue aggregates, payment refund action
/ratings	Ratings.jsx	Auth Admin	Quality assurance audit, star distribution, customer feedback comments
/support	SupportDashboard.jsx	Auth Admin	Central helpdesk queue, live two-way customer chat, ticket resolution
/translations	Translations.jsx	Auth Admin	Dynamic bilingual dictionary editor, search and inline modification

7.4 Screen-to-Feature Mapping

Feature Area	Resident Screen	Technician Screen	Admin Screen	Backend Service (.NET 8)
Authentication & Verification	R-01, R-02, R-03	T-01 (Login)	A-Auth	AuthService.cs / AuthController.cs
Compound & Category Setup	R-04	T-01	A-01, /categories, /compounds	CompoundController.cs
Order Dispatch & Bidding	R-05, R-06	T-01 (Radar, Bid)	A-02 (Board), /orders	OrderService.cs / OrderHub.cs
GPS Tracking & Navigation	R-07 (TrackingMap)	T-02 (TrackingMap)	/orders (Inspection)	LocationService.cs / LocationHub.cs
Job State Progression	R-07 (Timeline)	T-02 (Action CTA)	A-02 (Live Lanes)	OrderService.cs / OrderHub.cs
In-Order Messaging	R-07 (Chat Modal)	T-02 (Chat Modal)	—	ChatController.cs / OrderHub.cs
Payments & Invoicing	R-07 (Pay Modal)	—	/payments (Ledger & Refund)	PaymentService.cs / PaymentController.cs
Rating & Quality Control	R-07 (Rate Modal)	T-01 (Star Display)	/ratings (Reviews Table)	RatingService.cs / RatingController.cs
Customer Support Helpdesk	R-09 (Widget)	—	A-03 (Support Desk)	SupportTicketController.cs / SupportHub.cs
Bilingual Localization	R-08 (Lang Select)	/profile (Lang Select)	A-04 (Translations)	TranslationController.cs

Section 8 — Responsive & Interaction Behavior

8.1 Viewport Breakpoint Matrix

Breakpoint	Target Device	Layout Behavior
< 768px	Smartphones (Mobile Web / PWA)	Native full-screen view, bottom tab navigation bar, touch-friendly tap targets (minimum 44×44px)
768px – 1024px	Tablets	Resident/Tech apps center within framed smartphone shell. Admin dashboard wraps metric cards into 2 columns.
> 1024px	Desktop Laptops & Monitors	Admin dashboard displays full 4-column metric grid and side-by-side Kanban board. Resident/Tech apps remain centered in mobile preview shell for optimal ergonomics.

8.2 Interaction & State Behavior

State Type	Description	Implementation
Loading	Skeleton loaders during API fetch	Loading spinner inside buttons; skeleton cards in feed areas
Empty State	No active orders, no bids yet	Empty active orders state with visual prompt
Error State	API failure, 401, 422 errors	Red toast notifications, error banners, AlertModal dialogs
Success State	Order submitted, payment confirmed	Success popup modals, toast notifications
Validation	Form field errors	Per-input validation error indicators via AppInput component
Real-Time Sync	WebSocket disconnection	Auto-reconnects via <code>@microsoft/signalr</code> with exponential backoff
RTL Support	Arabic language selected	RTL toggle in profile screen; bilingual toggle (AR/EN) switches layout direction
Dark Mode	Admin Console	Full dark theme with <code>#0F172A</code> base and <code>#1A2235</code> card surfaces
Light Mode	Resident & Technician portals	Off-white <code>#F8FAFC</code> background with white containers